Unit

27 Alkanes and alkenes

Unit Key Concepts

- Cracking of petroleum fractions
- Important reactions of alkanes and alkenes

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

27.1 Risks and benefits of fossil fuels to the society and environment

Although fossil fuels are finite resources, they will continue to play an important role in global energy supply. Fractions of petroleum are also used as raw materials to produce different chemicals in the petrochemical industry.

However, there are risks associated with the production, transportation, storage and usage of fossil fuels.

Worksheet **Assessment Form**

A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.

Find & Share

Risks and benefits of using fossil fuels to the society and environment

Task

Your teacher will divide the class in groups. Each group should give a 4-minute presentation on the above topic. Support your presentation with various aids, such as diagrams, maps, etc.

Hints for the search

1 What useful chemicals are produced from the petrochemical industry?

2 At what rate are fossil fuels being consumed in the world?

3 What are the risks associated with the production, transportation, storage and usage of fossil fuels?

4 What are the safety precautions in using household fuels?

Refer to page T27.

Reference websites

1 You may start your search from the following websites:

- An overview of the petrochemical industry http://www.elmhurst.edu/~chm/vchembook/325petrochem.html
- An article on the environmental impacts of fossil fuel use http://www.ucsusa.org/clean_energy/our-energy-choices/coal-and-other-fossil-fuels/the-hidden-cost-of-fossil.html

2 Extend your search using search engines and appropriate keywords.

Benefits of using fossil fuels

Some benefits of using fossil fuels are as follows:

- All three types of fossil fuels: petroleum, coal and natural gas are readily available at a relatively cheap price.
- The production of electricity from fossil fuels is not dependent on the time of day, weather variations or seasonal effects.
- Petroleum and natural gas can be transported easily by pipeline, ship, road and rail.
- Petroleum can be used to power cars.
- Petroleum can be refined to provide raw materials for the production of other products, such as polymers.

N1 Refer to page T33.

Risks of using fossil fuels

Some risks of using fossil fuels are as follows:

- Carbon dioxide is produced when fossil fuels are burnt. It is a greenhouse gas.
- Sulphur dioxide is produced when coal is burnt. It contributes to acid rain.
- A coal plant can release dust and fly ash.
- *Exploration* and mining have a major impact on the environment.
- Underground coal mining is dangerous.
- *Petroleum spills* during drilling and transportation cause major damage to *ecosystems* both on land and at sea.

exploration 探測　　petroleum spill 石油泄漏　　ecosystem 生態系統



Fig. 27.1 This oil tanker broke open and spilt a large amount of oil



Fig. 27.2 A sea bird covered in petroleum

Spillage of petroleum — a major source of water pollution

Petroleum is transported all over the world by large tankers. When accidents happen to these tankers, petroleum will leak into the sea (Fig. 27.1). This causes serious pollution problems due to the following reasons:

- Petroleum is toxic.
- Petroleum is flammable.
- Petroleum clogs the feathers of sea birds and prevents them from flying or swimming (Fig. 27.2). Birds trying to clean their feathers with their beaks may be poisoned.
- Petroleum also interferes with the *insulation* provided by the feathers of sea birds. Birds may die of cold.
- Petroleum floats on the water and blocks the oxygen supply to water lives.

When a spillage of petroleum occurs, experts will assess the extent of the spillage. They then decide whether to allow the action of wind and wave to disperse the petroleum, or to use artificial methods. Detergents may be sprayed on the petroleum to break it up into small droplets (Fig. 27.3). Then water current can carry the petroleum away. However, these chemicals may also damage water lives.

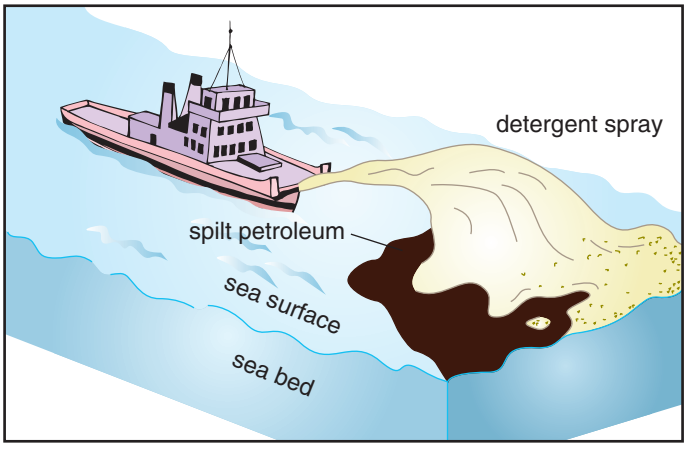


Fig. 27.3 Spraying detergent on the spilt petroleum

We will discuss this property of detergent in Topic 8 Chemistry of Carbon Compounds.

Do you know

BP oil spill

On April 20, 2010, a disaster hit the Gulf of Mexico Coast. British Petroleum's (BP) Deepwater Horizon oil rig exploded, spewing crude oil into the ocean.

From June 5 to June 14, BP collected 127 000 barrels of oil. However, it was believed that 60 000 barrels of oil were gushing into the Gulf per day.

The oil spill affected the shorelines and marshes. It could also disrupt the natural ecosystems.

insulation 保温作用

27.1
Investigating the properties of an alkane.

Video
Investigating the properties of an alkane

27.2 Important reactions of alkanes

Alkanes form a homologous series with the general formula C_nH_{2n+2}, where n = 1, 2, 3, 4, etc. Alkanes are chemically unreactive. They have strong carbon-carbon and carbon-hydrogen bonds. They do not react with acids, alkalis, oxidizing agents or reducing agents. However, they do burn and react with halogens under suitable conditions.

Combustion of alkanes

In a good supply of oxygen, alkanes undergo complete combustion to produce carbon dioxide and water, and release much heat. The following equation represents the complete combustion of methane, CH_4.

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$$ CE 2011 Paper 1 Q1(a)

The lighter alkanes are widely used as fuels. Natural gas (mainly methane) is used for cooking and heating in homes. Propane and butane are used as camping gas (Fig. 27.4).



Fig. 27.4 Propane and butane are used as camping gas

When the supply of oxygen is insufficient, incomplete combustion occurs. Carbon monoxide and even carbon (soot) are produced.

$$2C_8H_{18}(g) + 17O_2(g) \longrightarrow 16CO(g) + 18H_2O(l)$$

octane
an alkane found in petrol

Combustion of alkanes is a gas-phase reaction. Therefore liquid and solid alkanes must first be vaporized before they can be burnt. This is why less volatile alkanes burn less readily.

CE 2010 Paper 2 Q29

When the air hole of a Bunsen burner is fully open, enough oxygen is provided and the hydrocarbon fuel undergoes complete combustion. The flame obtained is non-luminous (Fig. 27.5).

CE 2011 Paper 1 Q1(a)

When the air hole is closed, the fuel undergoes incomplete combustion and a luminous flame is obtained (Fig. 27.6). The yellow flame is caused by the carbon particles it contains.



Fig. 27.5 When the air hole is opened, a non-luminous flame is produced



Fig. 27.6 When the air hole is closed, a luminous flame is produced

CE 2005 Paper 1 Q5(b)
CE 2011 Paper 1 Q6(a)(ii)

Example 27.1

Q Do you agree with the following statement? Explain your answer.

'The exhaust of diesel engine contains a higher concentration of particulates than that of petrol engine.'

A The carbon to hydrogen ratio of hydrocarbons in diesel is higher than that in petrol.

It is more difficult for diesel to undergo complete combustion.

Thus, burning diesel produces more particulates.

N2 Motor vehicles in some countries use gasohol (a mixture of ethanol and petrol) as fuel. Compared with petrol, gasehol undergoes complete combustion more readily.

Reference Website

Refer to the animation 'Substitution reaction':
http://www.youtube.com/watch?v=MvWQOYLm1f4
(accessed August 2014)

Reaction with halogens — substitution reactions

When alkanes react with halogens, substitution of hydrogen takes place. For the reactions to proceed, we must supply sufficient energy, either heat (about 300 °C) or ultraviolet light (from sunlight, for example).

When chlorine and methane gases are mixed in the presence of heat or ultraviolet light, chloromethanes and hydrogen chloride are produced. Notice that one or more hydrogen atom(s) in a methane molecule can be replaced by chlorine atom(s). This is an example of a **substitution reaction**. N3

When methane reacts with chlorine, the products chloromethane, dichloromethane, trichloromethane and tetrachloromethane are collectively known as chloromethanes.

$$CH_4(g) + Cl_2(g) \longrightarrow \underset{\text{chloromethane}}{CH_3Cl(g)} + HCl(g)$$ N4

CE 2010 Paper 2 Q24

$$CH_3Cl(g) + Cl_2(g) \longrightarrow \underset{\text{dichloromethane}}{CH_2Cl_2(l)} + HCl(g)$$

$$CH_2Cl_2(l) + Cl_2(g) \longrightarrow \underset{\text{trichloromethane}}{CHCl_3(l)} + HCl(g)$$

$$CHCl_3(l) + Cl_2(g) \longrightarrow \underset{\text{tetrachloromethane}}{CCl_4(l)} + HCl(g)$$

CE 2008 Paper 2 Q24
DSE 2014 Paper 1A Q10

The major product varies according to the proportion of chlorine to methane. If the amount of chlorine present is much larger than that of methane, the major product would be tetrachloromethane. However, if the amount of methane present initially is much larger than that of chlorine, chloromethane would be the major product.

Alkanes react with bromine in a similar manner (Fig. 27.7). However, N5 the reaction with bromine is slower than that with chlorine.

N3 The reaction of methane with chlorine is NOT suitable for the preparation of dichloromethane because a mixture of chloromethanes would form in the reaction.

N4 This is a redox reaction. The oxidation number of chlorine decreases from 0 to –1.

N5 Examination questions often ask about the reaction between bromine and alkane. The reddish brown reaction mixture changes to colourless gradually.

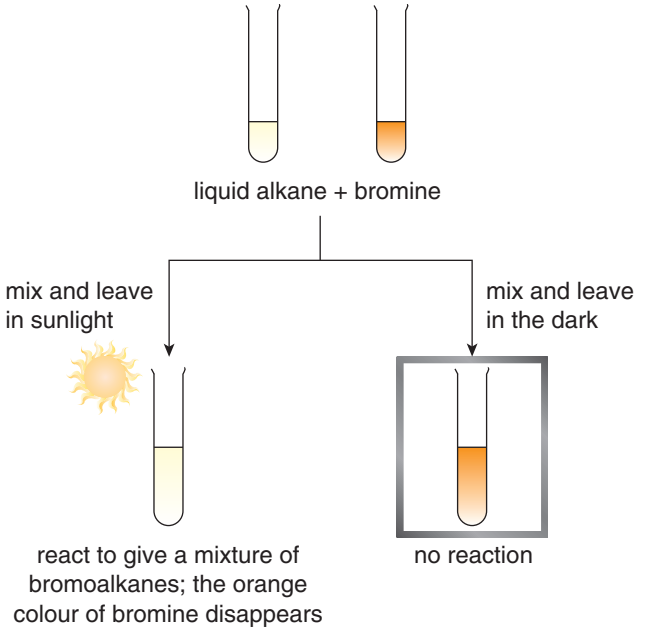


Fig. 27.7 A substitution reaction takes place between an alkane and bromine in sunlight but not in darkness

substitution reaction 取代反應

We will discuss haloalkanes in Topic 8 Chemistry of Carbon Compounds.

N6 The reaction between ethane and bromine gives a mixture of carbon compounds due to further substitution.

The products of reactions between alkanes and halogens are known as **haloalkanes**. A haloalkane is an alkane in which one or more hydrogen atom(s) has(have) been replaced by halogen atom(s). N6

✓ A substitution reaction is a reaction in which an atom (or group of atoms) in a molecule is replaced by another atom (or group of atoms).

Steps involved in the substitution of methane with chlorine

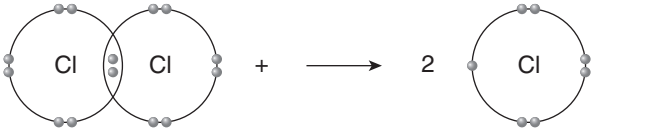
The following are the major steps involved in the reaction between methane and chlorine.

- **initiation** (how the reaction gets started);
- **propagation** (how the reaction keeps going);
- **termination** (how the reaction stops).

AL 2008 Paper 2 Q5(a)(v)

Initiation

Heat or ultraviolet light provides the energy required to split the chlorine molecule into atoms.



chlorine free radicals

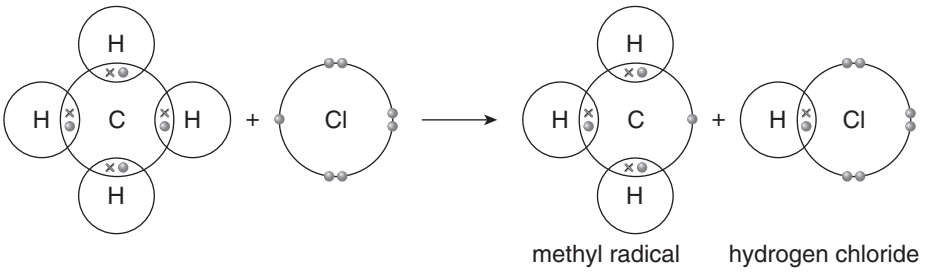
DSE 2012 Paper 1B Q15

The atoms are highly reactive because the unpaired electron has a strong tendency to pair up with another electron from another substance. These highly reactive atoms are called **free radicals**.

Propagation

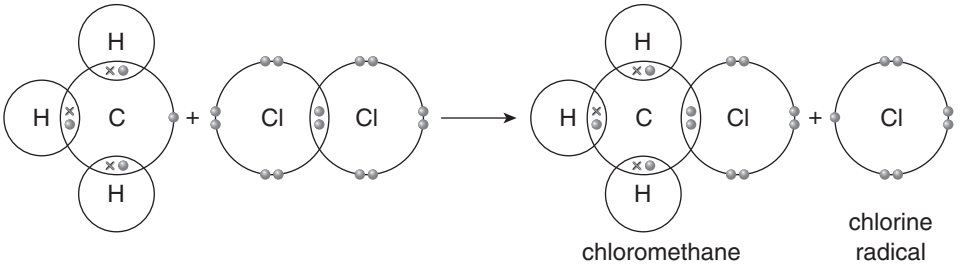
When a chlorine free radical collides with a methane molecule, it rips off one hydrogen atom to make a hydrogen chloride molecule. This process produces another radical, a methyl radical.

haloalkane 鹵烷　initiation 引發　propagation 傳播　termination 終止　free radical 自由基



DSE 2012 Paper 1B Q15

The methyl radical then reacts with another chlorine molecule to produce chloromethane and another chlorine radical.



The chlorine radical will continue to react with other molecules (e.g. CH_4, CH_3Cl, CH_2Cl_2, etc). Other products and free radicals will form during the process. These reactions enable a **chain reaction** to occur (Fig. 27.8).

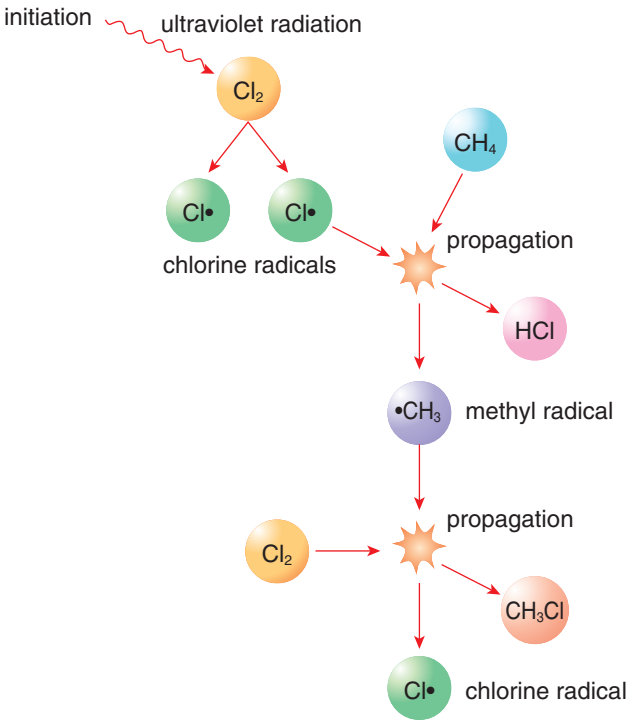
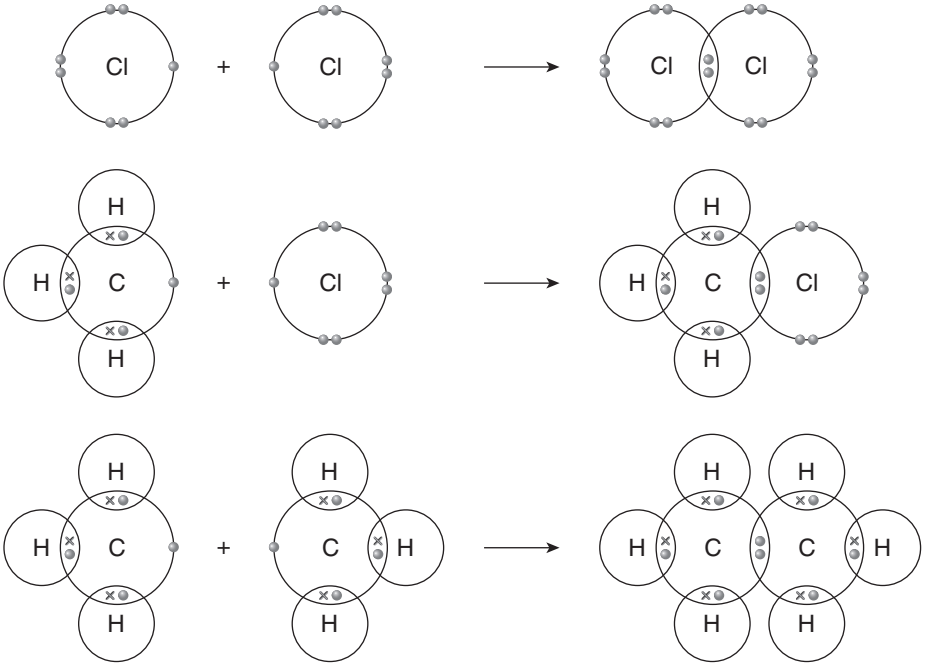


Fig. 27.8 The reaction of methane with chlorine is a free-radical chain reaction

chain reaction 連鎖反應

Termination

Whenever two free radicals meet, they react with each other to form a stable molecule. When no more free radicals are present to carry on the reaction sequence, the chain reaction stops.



Each chain may go through 100 to 10 000 cycles before termination occurs. The overall process of the reaction takes place in a very short period of time. Hence the reaction is explosive in nature.

Practice 27.1

1 For environmental reasons, the Hong Kong government has launched a plan for taxis and light buses to switch from using diesel to using liquefied petroleum gas (LPG). With reference to their chemical constituents, explain why LPG is a 'cleaner' fuel than diesel.

The carbon to hydrogen ratio of hydrocarbons in LPG is higher than that in petrol. LPG burns more completely, producing less particulate / carbon monoxide.

2 In an experiment, a mixture of bromine vapour and ethane was exposed to sunlight to produce a compound with the molecular formula C_2H_5Br.

a) State the expected observation in the experiment.
The mixture turns from reddish brown to colourless.

b) Name the type of reaction involved.
Substitution reaction

c) Suggest why this reaction is NOT an effective way for the preparation of C_2H_5Br.
The reaction gives a mixture of products ($C_2H_4Br_2$, $C_2H_3Br_3$, etc.)

27.3 Demand for various fractions of petroleum

There are many different kinds of petroleum. The exact composition of each kind of petroleum depends on the conditions in which it is formed. Table 27.1 shows the percentages of various fractions obtained from different kinds of petroleum.

The percentage of refinery gas is so small that it is not taken into account.

Table 27.1

Percentages of various fractions obtained from different kinds of petroleum

Type of petroleum	Percentage of fraction			
	Petrol	Kerosene	Diesel oil	Residue
North Sea	23	15	24	38
Arabian light	18	11.5	18	52.5
Arabian heavy	21	15	21	43
Iranian heavy	21	13	20	46

Fractions containing smaller hydrocarbon molecules are less viscous and easier to ignite. They also burn with a clearer flame. These properties make them more useful as fuels and more valuable than fractions that contain larger hydrocarbon molecules.

Fig. 27.9 shows an example of the supply and demand for various fractions obtained by fractional distillation of petroleum. The more useful fractions containing smaller hydrocarbon molecules, petrol for example, are in greater demand than the supply (Fig. 27.10). Fractions containing larger hydrocarbon molecules, residue for example, are less useful and the supply is greater than the demand.

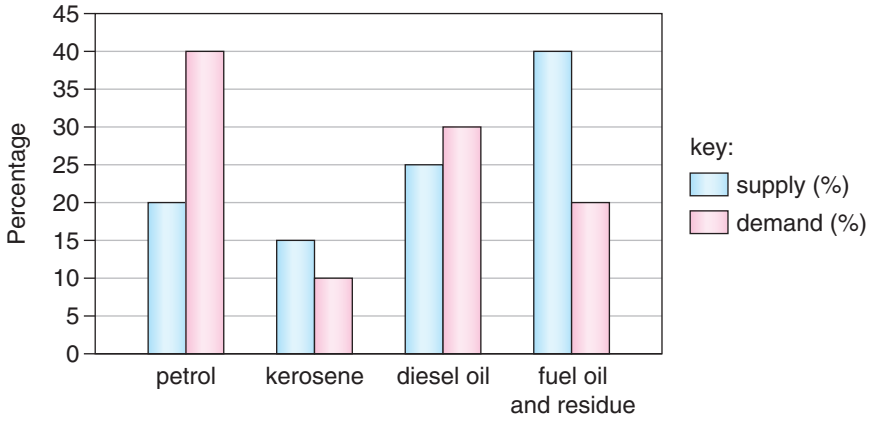


Fig. 27.9 Supply and demand for some fractions of petroleum



Fig. 27.10 Petrol is the main source of fuel for motor vehicles

27.4 Cracking

To solve the unbalanced supply and demand problem mentioned above, larger hydrocarbon molecules can be broken down into smaller ones by a process called **cracking**. N7

DSE 2012 Paper 1B Q2(a)(i)

Cracking can be achieved by using high pressure and temperature without a catalyst, or using lower pressure and temperature in the presence of a catalyst. The former way would be very expensive.

✓ Cracking is the breaking down of larger hydrocarbon molecules with heat or a catalyst to produce smaller hydrocarbon molecules.

CE 2007 Paper 2 Q49
CE 2009 Paper 1 Q5(a)

Both cracking and fractional distillation involve heating. However, cracking is a chemical change while fractional distillation is a physical change.

N7 Examination questions often ask about 'cracking'. Cracking is an endothermic reaction.

Catalytic cracking

The cracking process takes place in a steel vessel called a cracker (Fig. 27.11).

In the cracker, a high-boiling fraction is heated in the absence of air to vaporize the hydrocarbons. The vapour is then passed over a hot catalyst. The hydrocarbons are cracked. The large molecules split apart to form smaller ones. This method of cracking is called **catalytic cracking**.

Otherwise, the hydrocarbons will burn.



Fig. 27.11 A steel cracker

cracking 裂解　　catalytic cracking 催化裂解作用

Cracking of tetradecane ($C_{14}H_{30}$)

In general, the products obtained from cracking depend on the actual conditions used. Take the cracking of tetradecane ($C_{14}H_{30}$), an alkane found in kerosene, as an example. One typical cracking reaction is shown by the following equation:

$$\underset{\text{tetradecane}}{C_{14}H_{30}(l)} \longrightarrow \underset{\substack{\text{heptane}\\(\text{a component of petrol})}}{C_7H_{16}(l)} + \underset{\text{propene}}{C_3H_6(g)} + \underset{\text{ethene}}{2C_2H_4(g)}$$ N8

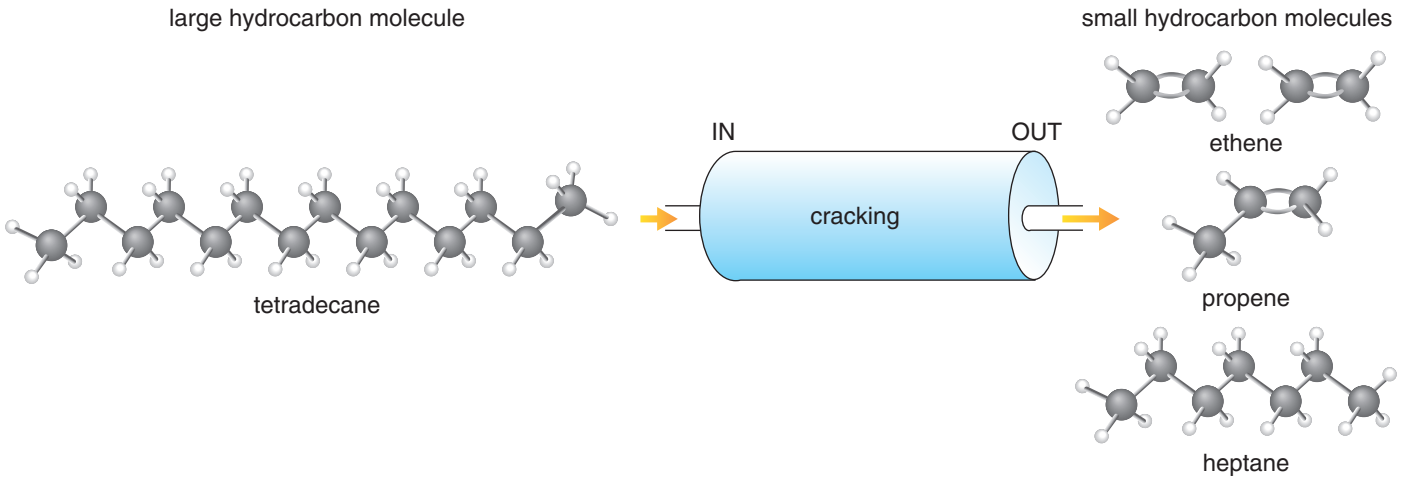


Fig. 27.12 The cracking of tetradecane (only some product molecules are shown)

Two different alkenes are produced — ethene and propene. Heptane, a component of petrol, is also formed (Fig. 27.12).

One other typical cracking reaction might be:

$$\underset{\text{tetradecane}}{C_{14}H_{30}(l)} \longrightarrow \underset{\text{dodecane}}{C_{12}H_{26}(l)} + \underset{\text{ethene}}{C_2H_4(g)}$$

This particular alkane may split apart in many other ways. The fraction being cracked also contains a mixture of hydrocarbons. At the end of the cracking process, the products obtained will include alkanes, alkenes, hydrogen and carbon. The mixture will be separated by fractional distillation.

N8 Examination questions often ask students to deduce molecular formulae of cracking products or write a chemical equation for the reaction.

Notice that the number of carbon / hydrogen atoms on the left-hand side and the right-hand side of the equation are equal.

e.g. $C_7H_{16}(l) \longrightarrow C_3H_8(g) + C_4H_8(g)$

Number of C atoms on left-hand side = 7
= 3 + 4
= number of C atoms on right-hand side

Number of H atoms on left-hand side = 16
= 8 + 8
= number of H atoms on right-hand side

Importance of cracking

Cracking is one of the most important reactions in an oil refinery.

DSE 2012 Paper 1B Q2(a)(ii)

Producing extra petrol

Cracking increases the production of petrol to meet demands.

DSE 2014 Paper 1B Q6(a)(ii)

As a source of alkenes

Cracking of alkanes produce alkenes. Alkenes (especially ethene and propene) are used to make a great variety of chemicals. Many of these N9 chemicals serve as raw materials in the production of other synthetic substances, such as polymers. Fig. 27.13 shows some substances produced from ethene. These substances will be discussed more in later units.

Polythene and polystyrene are polymers. They are formed by a process called polymerization. We will discuss polymers in Unit 28.

N9 Ethene is NOT manufactured directly by fractional distillation of petroleum.

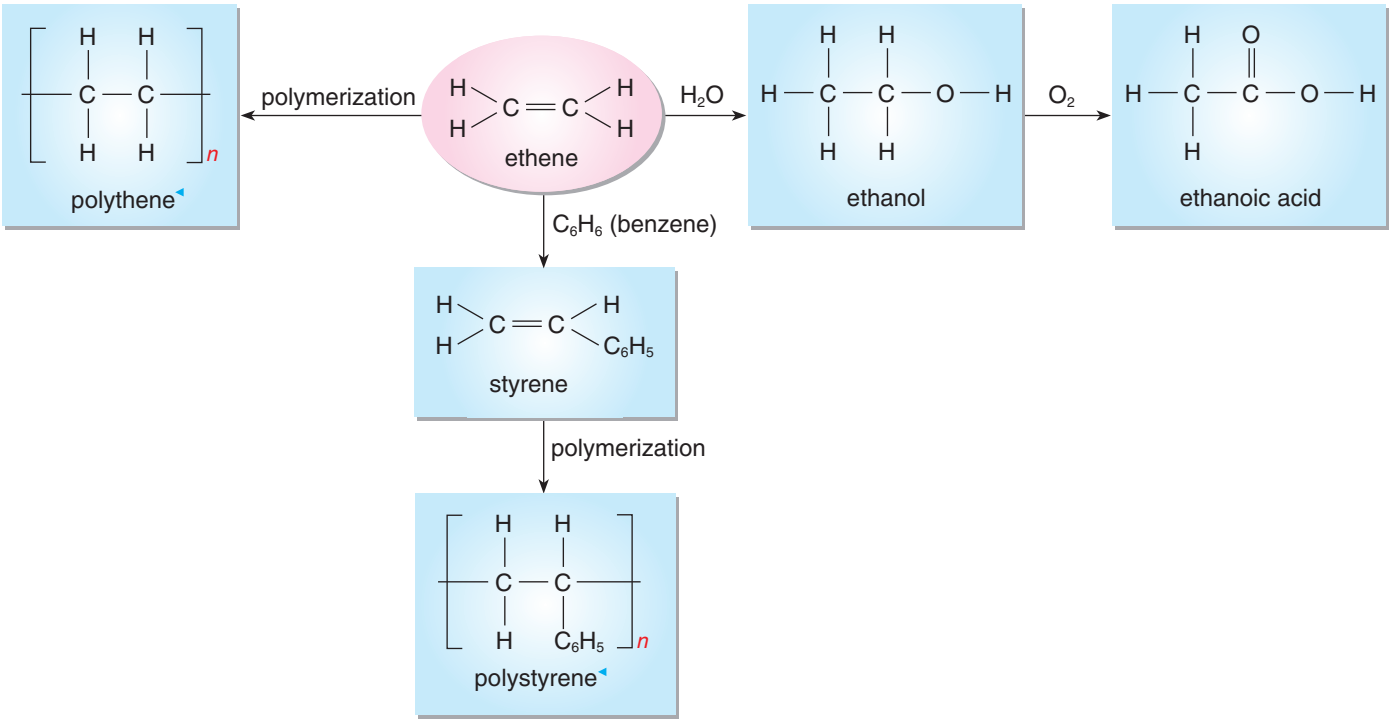


Fig. 27.13 Some substances produced from ethene

Video

Cracking of medicinal paraffin and testing the gaseous product

Reference Website

Refer to the video clip 'Cracking a hydrocarbon':
http://www.youtube.com/watch?v=ZYyKUePdC2Y
(accessed August 2014)

Vaporization of medicinal paraffin occurs when the mineral wool is heated. Cracking occurs when the vapour passes over the unglazed porcelain pieces.

27.2

Cracking of medicinal paraffin and testing the gaseous product.

Cracking of medicinal paraffin in the laboratory

We can demonstrate the cracking process in the laboratory by cracking **medicinal paraffin**. Fig. 27.14 shows the experimental set-up for the process. The gaseous product is collected over water.

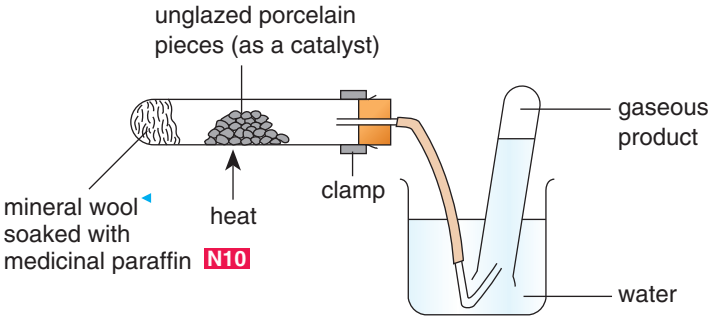


Fig. 27.14 Experimental set-up for the cracking of medicinal paraffin in the laboratory N11

N10 NO chemical reaction occurs at the mineral wool.

N11 Examination questions often ask about the laboratory set-up for cracking. Do NOT confuse this laboratory set-up with that for fractional distillation or reflux.

Practice 27.2

Hexane (C_6H_{14}) is a member of the homologous series of alkanes.

a) Name the raw material from which hexane is obtained.
Petroleum

b) One molecule of an alkane X can be cracked to form one molecule of hexane and two molecules of propene.

i) Deduce the molecular formula of X.
$C_{12}H_{26}$

ii) Write the chemical equation for the cracking reaction.
$C_{12}H_{26} \longrightarrow C_6H_{14} + 2C_3H_6$

iii) Describe how cracking is carried out.
Heat to vaporize alkane X.
Pass the vapour over a catalyst.

iv) Explain the main economic reason why alkanes are cracked.
To meet the large demand of lighter fractions, fractions with smaller number of carbon atoms per molecule.

27.5 The alkenes

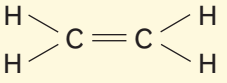
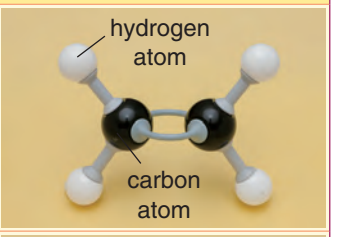
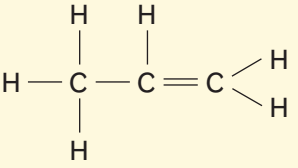
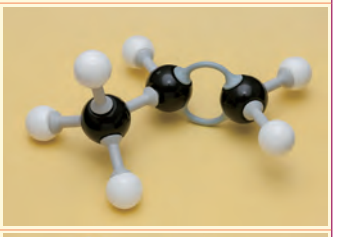
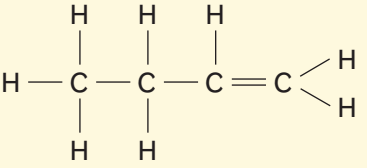
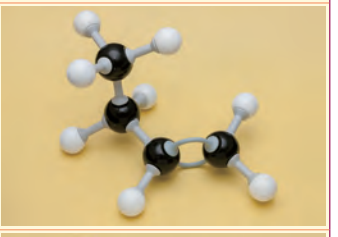
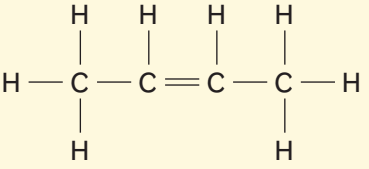
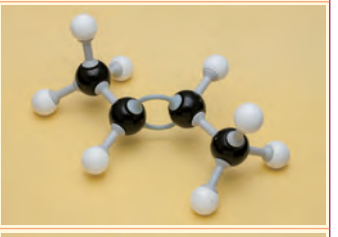
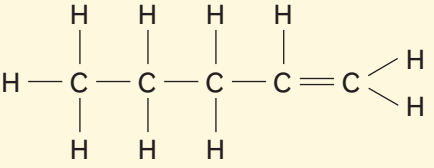
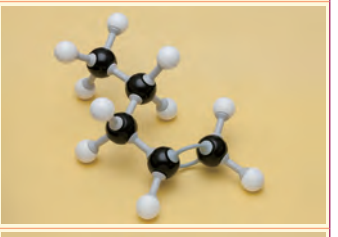
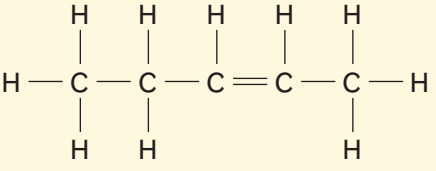
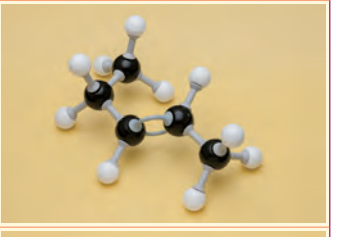
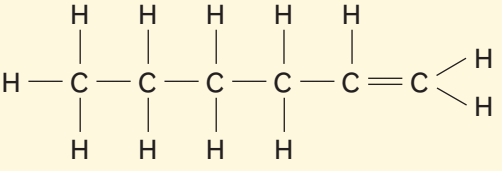
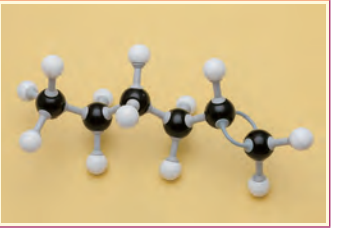
Alkenes are made from alkanes by the cracking process. Alkenes are unsaturated hydrocarbons with a carbon-carbon double bond $>C=C<$. They form a homologous series with the general formula C_nH_{2n}, where n = 2, 3, 4, etc.

medicinal paraffin 藥用石蠟

Table 27.2 shows the structural formulae and molecular models of some straight-chain alkenes.

Table 27.2

Structural formulae and molecular models of some straight-chain alkenes

Name	Molecular formula	Structural formula	Molecular model
Ethene	C_2H_4		
Propene	C_3H_6		
But-1-ene	C_4H_8		
But-2-ene	C_4H_8		
Pent-1-ene	C_5H_{10}		
Pent-2-ene	C_5H_{10}		
Hex-1-ene	C_6H_{12}		

27.6 Physical properties of alkenes

The melting and boiling points of alkenes are very close to those of the corresponding alkanes. It is because the members of both series are hydrocarbons.

There is also a gradual change of physical properties of alkenes as the length of the carbon chain in the molecules increases. Ethene, propene and butene are gases at room temperature, and the higher members are liquids and solids. Moreover, alkenes are insoluble in water.

CE 2011 Paper 2 Q10

Table 27.3 shows the physical properties of five straight-chain alkenes.

Table 27.3

Physical properties of five straight-chain alkenes

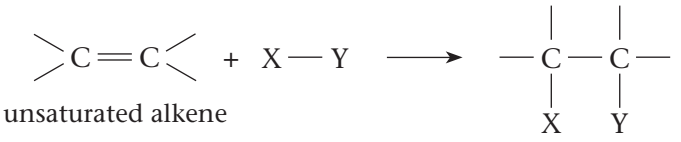
Name	Condensed structural formula	Melting point (°C)	Boiling point (°C)	State at room temperature and pressure
Ethene	$CH_2=CH_2$	–169	–104	gas
Propene	$CH_3CH=CH_2$	–185	–47	
But-1-ene	$CH_3CH_2CH=CH_2$	–185	–6	
Pent-1-ene	$CH_3(CH_2)_2CH=CH_2$	–138	30	liquid
Hex-1-ene	$CH_3(CH_2)_3CH=CH_2$	–140	63	

27.7 Important reactions of alkenes

Alkenes are much more reactive than alkanes. All alkenes have similar chemical properties because they all have the functional group $>C=C<$.

In many reactions of alkenes, the double bond is converted into single bonds. An atom or a group of atoms is added to each of the two carbon atoms with the double bond.

The general reaction is:



unsaturated alkene

a saturated compound

We will further discuss addition reactions of alkenes in Topic 8 Chemistry of Carbon Compounds.

N12 The reaction between an alkene and bromine / cold acidified dilute potassium permanganate solution is an 'addition reaction' NOT
✔
'additional reaction'.
✘

27.3
Investigating the properties of an alkene.

Video
Investigating the properties of an alkene

N13 Examination questions often ask about the reaction between bromine and alkene.

This is called an **addition reaction**. During an addition reaction, N12 an unsaturated alkene becomes saturated.

✓ An addition reaction is a reaction in which two or more molecules react to give a single product.

Only unsaturated compounds can undergo addition reactions.
CE 2006 Paper 2 Q49
CE 2009 Paper 2 Q39

Reaction with bromine

When an alkene is shaken with bromine dissolved in an organic N13 solvent, the orange colour of the bromine solution disappears quickly (Fig. 27.15).

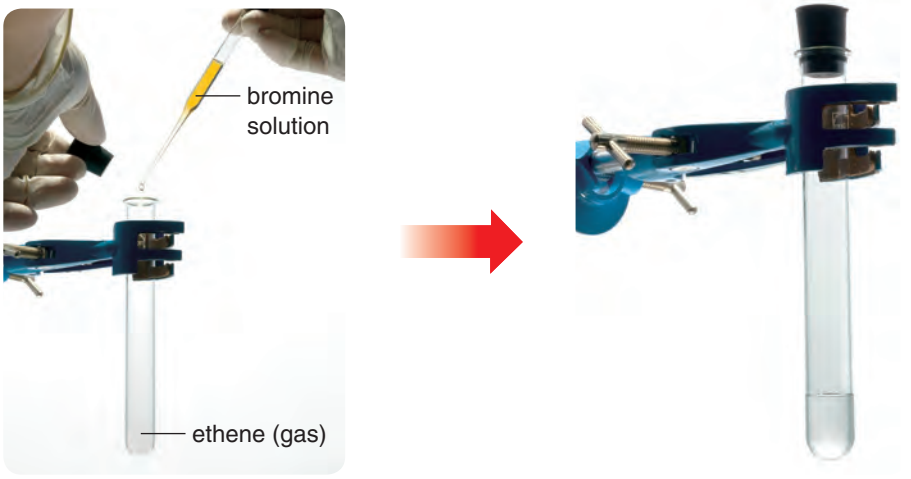
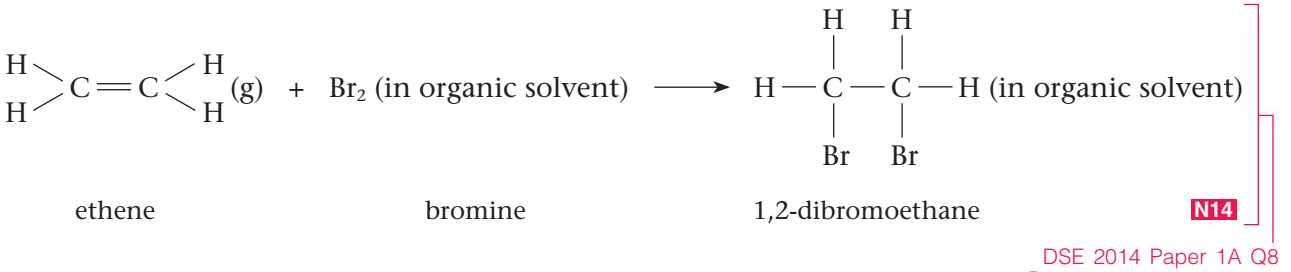


Fig. 27.15 Reaction of ethene with bromine (in organic solvent)

N14 Refer to page T33.

During the reaction, a bromine atom is added to each of the two carbon atoms with the double bond (Fig. 27.16). For example,



DSE 2014 Paper 1A Q8

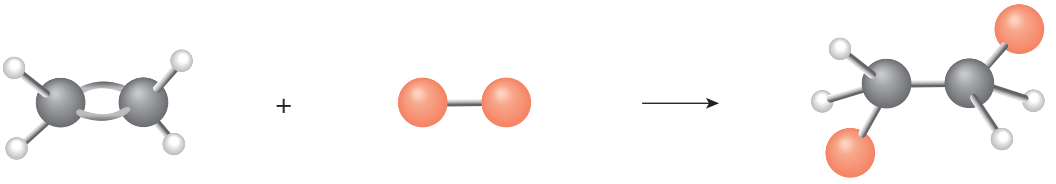
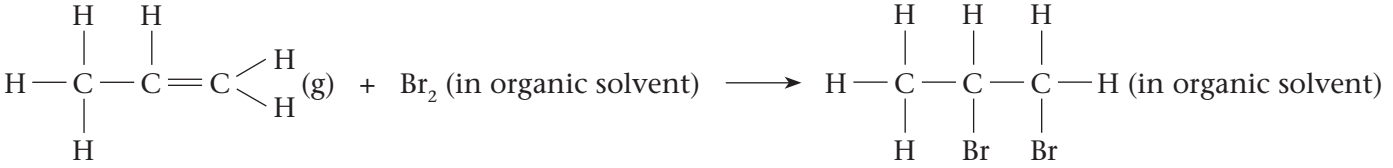


Fig. 27.16 The reaction between ethene and bromine is an addition reaction

addition reaction 加成反應

The reaction between ethene and bromine is an addition reaction. The product 1,2-dibromoethane is a colourless liquid. CE 2006 Paper 2 Q16 CE 2011 Paper 2 Q13

Propene also reacts with bromine to give 1,2-dibromopropane.



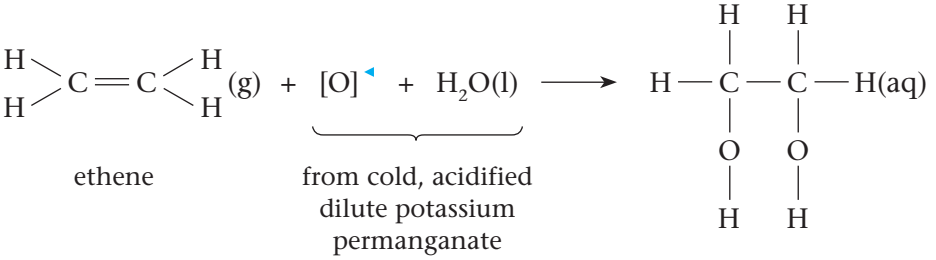
propene bromine 1,2-dibromopropane

The vigour of the addition reactions with the halogens decreases down the halogen group. Reactions between alkenes and iodine are relatively slow.

Reaction with cold acidified dilute potassium permanganate solution

CE 2007 Paper 2 Q27 CE 2008 Paper 2 Q13 CE 2009 Paper 2 Q24 CE 2010 Paper 2 Q38

When an alkene is shaken with a few drops of cold acidified dilute potassium permanganate solution, the purple solution will become colourless quickly (Fig. 27.17). In this reaction, one hydroxyl group (–O–H) is added to each of the two carbon atoms with the double bond to form a diol. For example,



ethene from cold, acidified dilute potassium permanganate solution ethane-1,2-diol

Acidified dilute potassium permanganate solution is a powerful oxidizing agent and it will cause further oxidation of the product ethane-1,2-diol. It requires cold and alkaline dilute potassium permanganate solution to produce ethane-1,2-diol.

[O] represents an oxidizing agent. In the reaction between an alkene and cold acidified dilute potassium permanganate solution, the oxidation number of manganese changes from +7 to +2.

DSE 2012 Paper 1A Q22

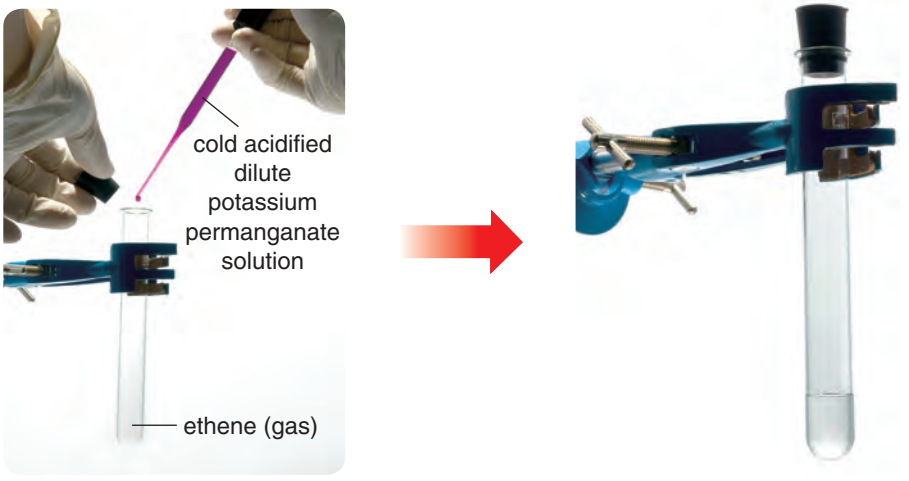
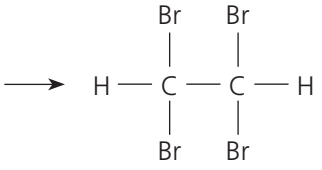


Fig. 27.17 Reaction of ethene with cold acidified dilute potassium permanganate solution

Ethyne (HC≡CH) is an unsaturated hydrocarbon. It can also decolourize the orange solution of bromine dissolved in an organic solvent quickly.

$$H{-}C{\equiv}C{-}H(g) + 2Br_2 \text{ (in organic solvent)} \longrightarrow$$



(in organic solvent)

27.4

Distinguishing between an alkane and an alkene. N16

N15 Refer to page T33.

N16 Refer to page T34.

Reference Website

Refer to the video 'Testing for unsaturated hydrocarbons with bromine': http://www.youtube.com/watch?feature=player_embedded&v=PE1CDR1S5pk (accessed August 2014)

Test for unsaturation

DSE 2012 Paper 1B Q2(c)(ii)
DSE 2014 Paper 1B Q3(a)

Two common tests for alkenes are:

1 The orange solution of bromine dissolved in an organic solvent becomes colourless quickly when shaken with an alkene (Fig. 27.18). N15

2 The purple solution of cold acidified dilute potassium permanganate becomes colourless quickly when shaken with an alkene.

However, these tests also apply to other unsaturated compounds. Hence these tests in fact show the existence of an unsaturated compound rather than just an alkene.

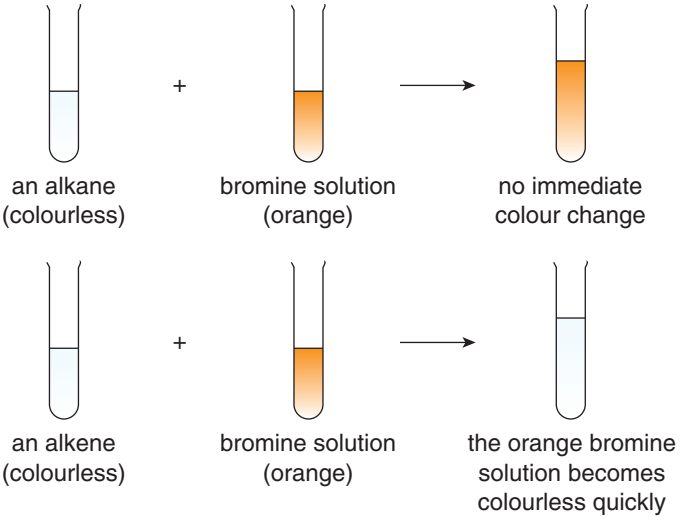


Fig. 27.18 The orange solution of bromine dissolved in an organic solvent can be used to distinguish between an alkane and an alkene

Example 27.2

Q Dodecane ($C_{12}H_{26}$) can be cracked into ethene and a straight-chain alkane X such that the mole ratio of ethene : straight-chain alkane X is 2 : 1.

a) Write a chemical equation for this reaction.

b) A student performed an experiment to crack dodecane and collected the gaseous products by using a boiling tube.

i) The student added a few drops of bromine dissolved in an organic solvent into the boiling tube containing the gaseous products. The orange colour of the bromine solution disappeared immediately. Why?

ii) The student then dropped more bromine solution into the boiling tube until the orange colour of the bromine solution persisted. After about 10 minutes, the orange colour disappeared. Why?

Continued on next page

A a) $C_{12}H_{26}(l) \longrightarrow 2C_2H_4(g) + C_8H_{18}(l)$

CE 2011 Paper 1 Q12

b) i) The products of cracking contained alkenes / unsaturated hydrocarbons which decolourized the bromine solution quickly by addition reaction.

ii) The products of cracking contained alkanes / saturated hydrocarbons which decolourized the bromine solution slowly by substitution reaction.

Example 27.3

Q Compound X contains carbon and hydrogen only. The relative molecular mass of X is 56.0. Complete combustion of 1.12 g of X gives 3.52 g of carbon dioxide.

a) Deduce the molecular formula of X.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

b) Bromine becomes colourless quickly when several drops of bromine dissolved in an organic solvent are added to X.

i) Draw TWO possible structures of X. Explain your answer.

ii) Give the IUPAC name of the compound represented by each structure drawn in (i).

A a) Mass of carbon in 1.12 g of X

$= 3.52 \text{ g} \times \frac{12.0}{44.0} = 0.960 \text{ g}$

Mass of hydrogen in 1.12 g of X
= (1.12 – 0.960) g = 0.160 g

	Carbon	Hydrogen
Mass of element in the compound	0.960 g	0.160 g
Number of moles of atoms that combine	$\frac{0.960 \text{ g}}{12.0 \text{ g mol}^{-1}}$ = 0.0800 mol	$\frac{0.160 \text{ g}}{1.00 \text{ g mol}^{-1}}$ = 0.160 mol
Simplest ratio of atoms	$\frac{0.0800 \text{ mol}}{0.0800 \text{ mol}}$ = 1.00	$\frac{0.160 \text{ mol}}{0.0800 \text{ mol}}$ = 2.00

∴ the empirical formula of X is CH_2.

Let $(CH_2)_n$ be the molecular formula of X.

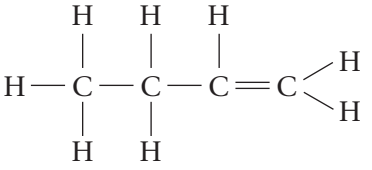
Relative molecular mass of X
$= n(12.0 + 2 \times 1.0)$
$= 14n$

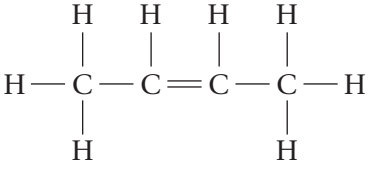
i.e. $14n = 56.0$
$n = 4$

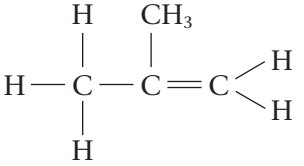
∴ the molecular formula of X is C_4H_8.

b) X contains a C=C bond because it can decolourize bromine dissolved in an organic solvent quickly.

Any two of the following structures:


but-1-ene


but-2-ene


methylpropene

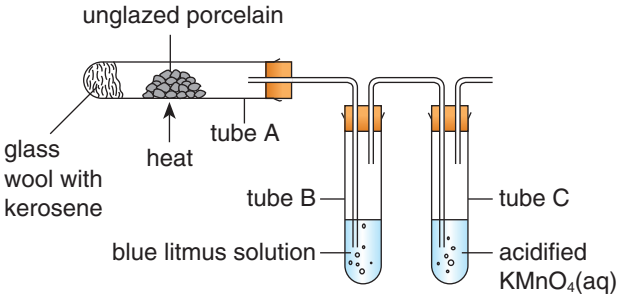
Practice 27.3 Refer to page T34.

DSE 2012 Paper 1A Q17

1 Both propene and but-1-ene are members of the alkene homologous series.

Using propene and but-1-ene as examples, illustrate TWO characteristics of the members of a homologous series.

2 Consider the following set-up of an experiment.



a) What is the chemical process that occurs in the experiment?

b) What is the function of the unglazed porcelain?

c) Will there be a colour change in each of the following solutions? Explain your answer.

i) Solution in tube B

ii) Solution in tube C

3 Compound Z contains carbon, hydrogen and oxygen only. The relative molecular mass of Z is 60.0. 1.50 g of Z contain 0.900 g of carbon and 0.200 g of hydrogen.

a) Deduce the molecular formula of Z.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

b) i) Draw TWO possible structures of Z.

ii) Give the IUPAC name of the compound represented by each structure drawn in (i).

Animation
Wind power as an alternative source of energy

Video
Wind power station in Hong Kong

27.8 Wind power — an alternative source of energy

Fossil fuels will run out one day. Many countries are looking for alternatives to fossil fuels as energy sources.

Wind power is the world's fastest growing energy source for electricity generation. The first commercial-scale wind power station in Hong Kong was formally commissioned in 2006 (Fig. 27.19).

Wind energy is a clean and **renewable energy source**. Wind power N17 has become one of the most promising renewable energy sources. Over the last decade, the wind industry has seen tremendous growth, and wind farms are popping up all over the world. It is currently the renewable energy source that can best compete with fossil fuel power plants — in some regions around the world, the clean and safe energy it generates is already just as cheap as that produced by coal.

CE 2009 Paper 2 Q16



Fig. 27.19 The first commercial wind power station on Lamma Island

N17 Refer to page T36.

renewable energy source 可再生能源

Do you know

Lamma Winds

The wind power station on Lamma Island, known as Lamma Winds, was constructed by the Hongkong Electric Co. Ltd.. Wind power can now be harnessed on a commercial scale to supply electricity to local businesses and residents. Through this project, the Hongkong Electric Co. Ltd. hopes to explore wind power as a potential supplement to fossil fuels for electricity generation.

Lamma Winds has also made a major contribution to Hong Kong society's awareness of the entire sustainable energy issue.

Chemistry Magazine

The fossil fuel dilemma

CE 2009 Paper 2 Q16

Petroleum, coal and natural gas are non-renewable resources. They cannot be replaced when they have all been used up. Predictions vary as to how long they will last. However, it seems likely that at the current rate of use, petroleum and natural gas will run out within this century. Coal is likely to last much longer than petroleum and natural gas.

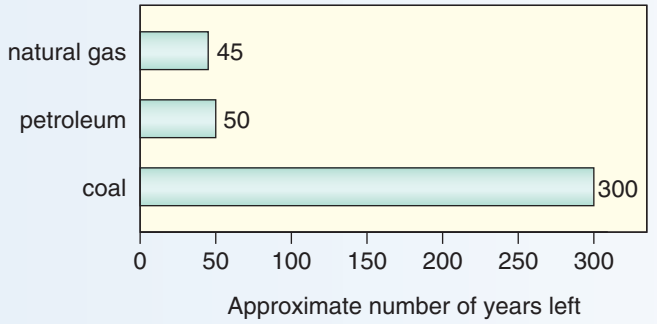


Fig. 27.20 Predictions of how long the fossil fuels will last at the current rate of use

So, should we increase our efforts to develop renewable forms of energy such as wind energy? Should we put greater emphasis on nuclear power? Or should we plan to make much greater use of coal? Perhaps we should do all three. Solving this dilemma is likely to depend as much on political decisions as scientific ones. What would you recommend?

Questions

1 Discuss the pros and cons of using coal for generating electricity.

2 Discuss the pros and cons of using wind power as an alternative energy source in Hong Kong.

Refer to page T37.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

Have you mastered?

Key terms

substitution reaction 取代反應 77 haloalkane 鹵烷 78 initiation 引發 78

propagation 傳播 78 termination 終止 78 free radical 自由基 78

chain reaction 連鎖反應 79 cracking 裂解 82 catalytic cracking 催化裂解作用 82

medicinal paraffin 藥用石蠟 85 addition reaction 加成反應 88

renewable energy source 可再生能源 92

Checklist

After studying this unit, you should be able to

- ☐ discuss the risks and benefits of using fossil fuels to the society and the environment;
- ☐ describe the following reactions of alkanes:
  - combustion;
  - substitution reactions with chlorine and bromine;
- ☐ describe the steps involved in the monosubstitution of methane with chlorine using electron diagrams;
- ☐ recognize the demand for the various petroleum fractions;
- ☐ recognize that cracking is a process to obtain smaller molecules including alkanes and alkenes;
- ☐ describe how to carry out laboratory cracking of a petroleum fraction;
- ☐ explain the importance of cracking in the petroleum industry;
- ☐ describe the reactions of alkenes with:
  - bromine dissolved in an organic solvent;
  - cold acidified dilute potassium permanganate solution;

☐ describe chemical tests for distinguishing between an alkane and an alkene;

☐ discuss the pros and cons of using wind power in Hong Kong.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 The following table summarizes two important reactions of alkanes.

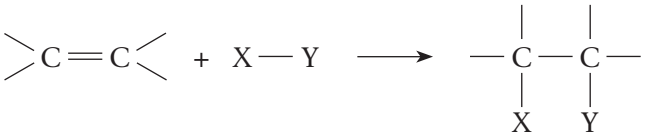
Reaction	Examples
1 Combustion — in a good supply of oxygen, alkanes burn completely to form carbon dioxide and water	Complete combustion: $2C_8H_{18}(g) + 25O_2(g) \longrightarrow 16CO_2(g) + 18H_2O(l)$ Incomplete combustion: $2C_8H_{18}(g) + 17O_2(g) \longrightarrow 16CO(g) + 18H_2O(l)$ $2C_8H_{18}(g) + 9O_2(g) \longrightarrow 16C(s) + 18H_2O(l)$
2 Substitution reaction with halogens — formation of haloalkanes	$CH_4(g) + Cl_2(g) \longrightarrow CH_3Cl(g) + HCl(g)$ $CH_3Cl(g) + Cl_2(g) \longrightarrow CH_2Cl_2(l) + HCl(g)$ $CH_2Cl_2(l) + Cl_2(g) \longrightarrow CHCl_3(l) + HCl(g)$ $CHCl_3(l) + Cl_2(g) \longrightarrow CCl_4(l) + HCl(g)$

2 a) The melting and boiling points of alkenes are very close to those of the corresponding alkanes.

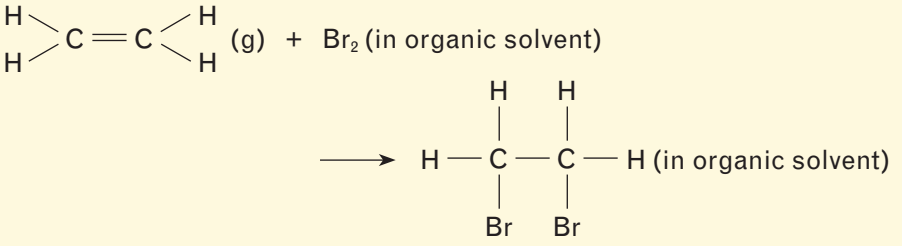
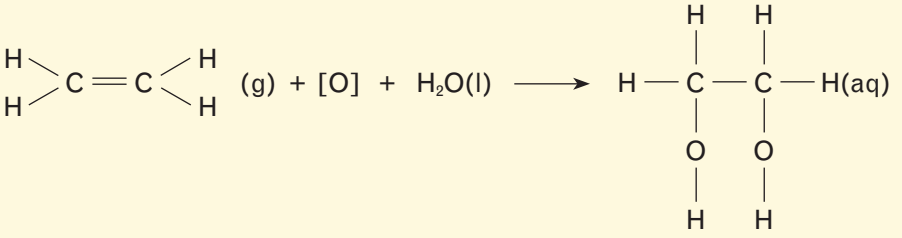
Name	Condensed structural formula	Melting point (°C)	Boiling point (°C)	State at room temperature and pressure
Ethene	$CH_2=CH_2$	–169	–104	gas
Propene	$CH_3CH=CH_2$	–185	–47	
But-1-ene	$CH_3CH_2CH=CH_2$	–185	–6	
Pent-1-ene	$CH_3(CH_2)_2CH=CH_2$	–138	30	liquid
Hex-1-ene	$CH_3(CH_2)_3CH=CH_2$	–140	63	

b) Alkenes are insoluble in water.

3 Alkenes are unsaturated hydrocarbons. They undergo addition reactions.



4 The following table summarizes two addition reactions of alkenes. These reactions can be used as tests for unsaturated compounds.

Reaction	Example
With bromine dissolved in an organic solvent — the orange solution of bromine becomes colourless quickly when shaken with an alkene	
With cold acidified dilute potassium permanganate solution — the purple solution of potassium permanganate becomes colourless quickly when shaken with an alkene	

5 The following diagram summarizes some important information about cracking.

Cracking

The breaking down of larger hydrocarbon molecules with heat or a catalyst to produce smaller hydrocarbon molecules

Importance of cracking

- Producing extra petrol to meet demands
- As a source of alkenes to make a great variety of chemicals

Examples:

$C_{14}H_{30}(l) \longrightarrow C_7H_{16}(l) + C_3H_6(g) + 2C_2H_4(g)$
tetradecane — heptane (a component of petrol) — propene — ethene

$C_{14}H_{30}(l) \longrightarrow C_{12}H_{26}(l) + C_2H_4(g)$
tetradecane — dodecane — ethene

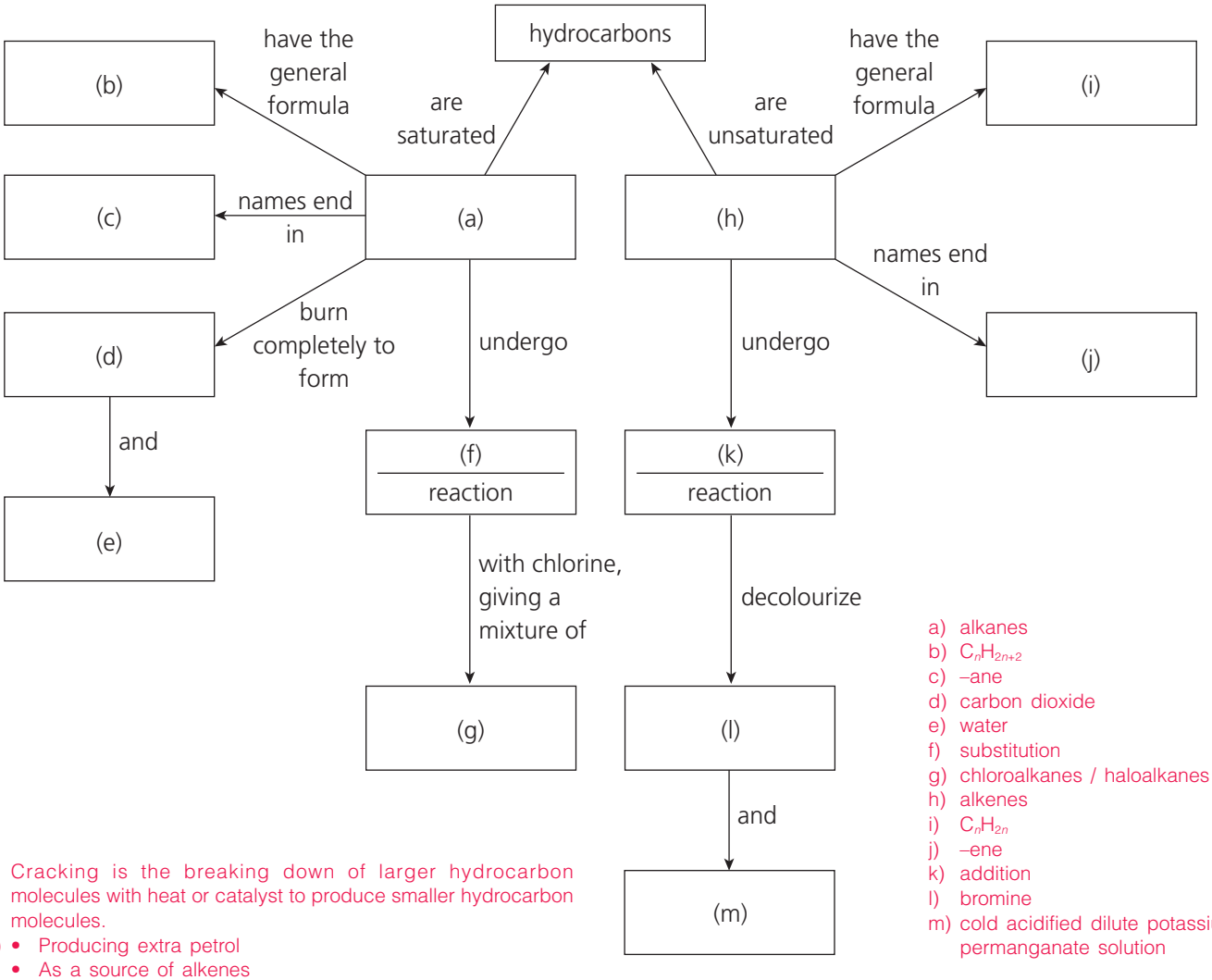
Have you mastered?

Unit Exercise

Note: The symbol 🌶 indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.



a) alkanes
b) C_nH_{2n+2}
c) –ane
d) carbon dioxide
e) water
f) substitution
g) chloroalkanes / haloalkanes
h) alkenes
i) C_nH_{2n}
j) –ene
k) addition
l) bromine
m) cold acidified dilute potassium permanganate solution

2 a) Cracking is the breaking down of larger hydrocarbon molecules with heat or catalyst to produce smaller hydrocarbon molecules.
b) • Producing extra petrol
• As a source of alkenes
c) $C_8H_{18}(l) \longrightarrow C_5H_{12}(l) + C_3H_6(g)$

2 a) Explain the term 'cracking'.

b) Why is cracking an important process in an oil refinery?

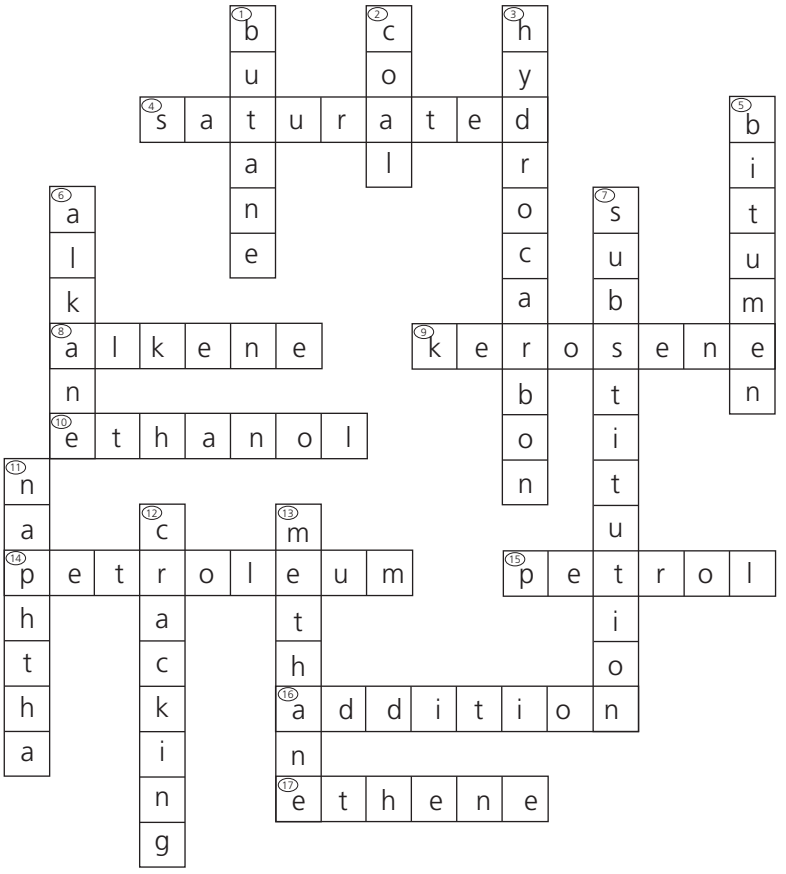
c) Write an equation for the cracking of octane (C_8H_{18}) in which one of the products is pentane (C_5H_{12}).

3 Describe a simple chemical test to distinguish an unsaturated hydrocarbon from a saturated one.

Any one of the following:
- Shake each hydrocarbon with an orange solution of bromine dissolved in an organic solvent.
  The orange solution becomes colourless quickly when shaken with an unsaturated hydrocarbon.
  There is no immediate observable change for a saturated hydrocarbon.
- Shake each hydrocarbon with cold acidified dilute potassium permanganate solution.
  The purple solution becomes colourless quickly when shaken with an unsaturated hydrocarbon.
  There is no observable change for a saturated hydrocarbon.

4 Write clues for the following crossword.

Refer to page T38.



Part II Multiple choice questions

5 The equation below represents the complete combustion of organic compound X:

$2X + 9O_2 \longrightarrow 6CO_2 + 8H_2O$

What is X?

A C_3H_6
B C_3H_8
C C_3H_6O
(D) C_3H_8O

(HKCEE, Paper 2, 2011, 18)

Directions: Questions 6 and 7 refer to the information below.

The equations below show some of the processes that occur when methane and chlorine react.

A $Cl_2(g) \longrightarrow 2Cl\bullet(g)$
B $Cl\bullet(g) + CH_4(g) \longrightarrow CH_3\bullet(g) + HCl(g)$
C $CH_3\bullet(g) + CH_3\bullet(g) \longrightarrow C_2H_6(g)$
D $Cl_2(g) + CH_4(g) \longrightarrow CH_3Cl(g) + HCl(g)$

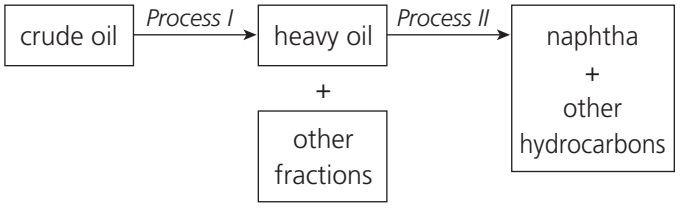
6 Which equation shows a propagation step? B

7 Which equation shows an initiation step? A

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2010, 14(a)–(b))

Unit Exercise

8 Consider the industrial processes as shown below:



Which of the following combinations is correct?

	Process I	*Process II*
A	is a chemical change.	is a physical change.
B	is a chemical change.	is a chemical change.
C	is a physical change.	is a physical change.
(D)	is a physical change.	is a chemical change.

(HKCEE, Paper 2, 2010, 2)

9 The equation below represents the cracking of a hydrocarbon:

$C_{22}H_{46} \longrightarrow C_{14}H_{30} + 2X$

What is the chemical formula of compound X?

A C_3H_6
(B) C_4H_8
C C_8H_{16}
D $C_{14}H_{28}$

(HKDSE, Practice paper 1A, 2012, 11)

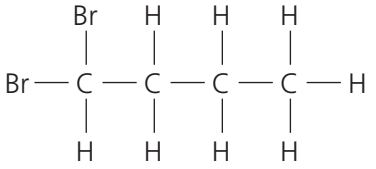
10 Which of the following mixtures could NOT form when octane, C_8H_{18}, is cracked?

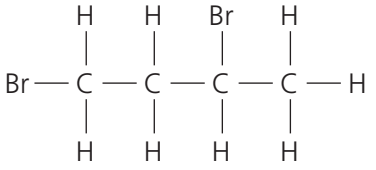
A propane + pentene
B butane + butene
C pentane + propene
(D) heptane + ethene

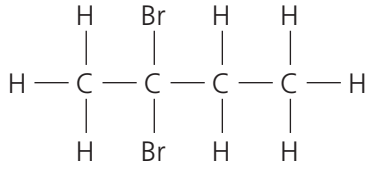
(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2010, 13)

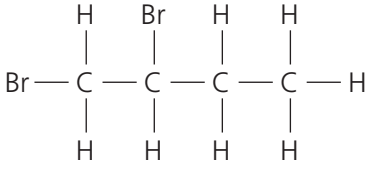
One possible cracking reaction of C_8H_{18} is shown below.
$C_8H_{18}(l) \longrightarrow C_6H_{14}(l) + C_2H_4(g)$
The cracking of C_8H_{18} could NOT form heptane and ethene.

11 Which of the following compounds is formed from the reaction of but-1-ene with bromine?

A


B


C


(D)


12 Which of the following statements concerning the reaction between acidified potassium permanganate solution and excess propene is INCORRECT?

A The oxidation number of manganese changes from +7 to +2.
B The reaction occurred is an addition reaction.
C The acidified potassium permanganate solution is decolourized.
(D) The structure of the organic product is $CH_2(OH)CH_2CH_2OH$.

(HKCEE, Paper 2, 2008, 13)

11 

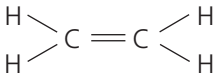
13 Ethane reacts with chlorine under suitable conditions. Which of the following statements concerning this reaction is / are correct?

(1) The reaction occurs readily in the dark.
(2) The reaction is initiated by the formation of chlorine free radicals.
(3) The reaction gives a mixture of carbon compounds.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(1) Ethane does NOT react with chlorine in the dark.

14 Both ethene and ethyne are gaseous hydrocarbons. Their structures are shown below:



ethene



ethyne

Which of the following statements concerning ethene and ethyne are correct?

(1) Both are members of the same homologous series.
(2) Both can decolourize cold acidified dilute potassium permanganate solution.
(3) Ethyne burns with a sootier flame than ethene.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

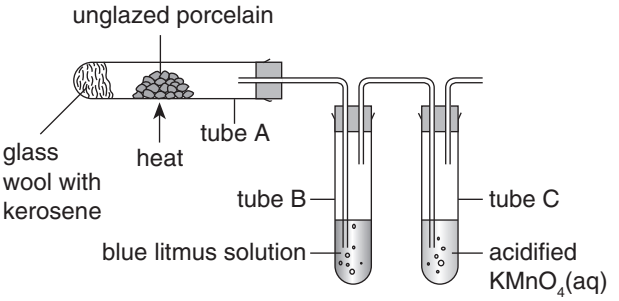
15 Which of the following statements about but-2-ene are correct?

(1) It has the same molecular formula as but-1-ene.
(2) It is soluble in water.
(3) It can decolourize bromine dissolved in an organic solvent.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(1) Both but-2-ene and but-1-ene have the molecular formula C_4H_8.
(2) But-2-ene is insoluble in water.

16 The diagram below shows the set-up of an experiment:



The unglazed porcelain in tube A is strongly heated and the glass wool is occasionally heated. Which of the following statements is / are correct?

(1) A chemical reaction occurs at the glass wool.
(2) There is NO colour change in the solution in tube B.
(3) There is NO colour change in the solution in tube C.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKDSE, Paper 1A, 2012, 17)

17 Which of the following experiments would give a reaction?

(1) Mixing ethane and bromine vapour in the dark.
(2) Mixing ethene and bromine dissolved in an organic solvent in the dark.
(3) Mixing ethene and acidified dilute potassium dichromate solution under sunlight.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

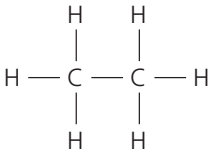
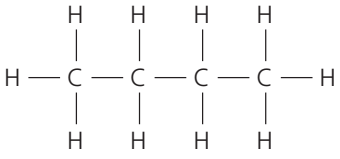
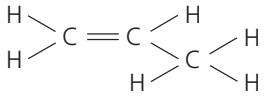
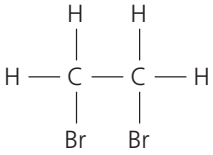
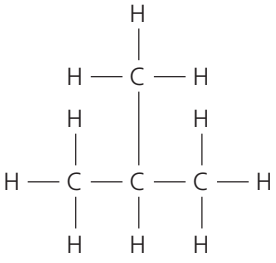
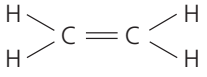
14 (1) Ethene and ethyne contain different functional groups. They belong to different homologous series.
(2) Both ethene and ethyne are unsaturated hydrocarbons. They can decolourize cold acidified dilute potassium permanganate solution.
(3) Ethyne has a higher carbon to hydrogen ratio than ethene. It is more difficult for ethyne to undergo complete combustion. Thus, it burns with a sootier flame than ethene.

17 (1) Ethane does NOT react with bromine in the dark.
(2) Ethene undergoes an addition reaction with bromine in the dark.
(3) There is NO reaction between ethene and acidified dilute potassium dichromate solution.

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

18 The table shows the structures of some carbon compounds.

 A	 B
 C	 D
 E	 F

a) Explain why compound C is NOT a saturated hydrocarbon.

b) Explain why compound D is NOT a hydrocarbon.

c) Name and give the general formula of the homologous series to which compound E belongs.

d) What colour change is seen when bromine dissolved in an organic solvent is added to compound F?

(Edexcel IGCSE (Higher Tier), Chemistry, Paper 2H, Nov. 2009, 5(a)–(b), (e)–(f))

18 a) Compound C contains a carbon-carbon double bond.
b) Compound D contains bromine atoms, not only carbon and hydrogen atoms.
c) Name of homologous series: alkane
General formula: C_nH_{2n+2}
d) The orange solution becomes colourless quickly.

19 Alkanes are saturated hydrocarbons. The first four straight-chain alkanes are shown below.

Refer to page T38.

methane	CH_4
ethane	CH_3CH_3
propane	$CH_3CH_2CH_3$
butane	$CH_3CH_2CH_2CH_3$

a) i) State what is meant by the term 'saturated hydrocarbon'.

ii) Give the general formula for the alkanes.

iii) Give the molecular formula for hexane, the sixth member of the series.

b) i) Write an equation for the complete combustion of butane.

ii) Write an equation for the incomplete combustion of butane to produce carbon monoxide and water.

iii) Under what conditions would you expect incomplete combustion to occur?

c) A mixture of butane and bromine vapour is exposed to diffused sunlight.

i) State the expected observation.

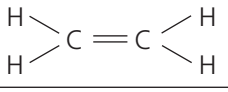
ii) Write a chemical equation for the reaction.

iii) Name the type of reaction that takes place.

20 A non-luminous flame is obtained when the air hole of a Bunsen burner is fully open. Methane is one of the components of the gaseous fuel used in the Bunsen burner. With reference to methane only and aided by a chemical equation, explain why the flame obtained is non-luminous.

(HKCEE, Paper 1, 2011, 1(a))

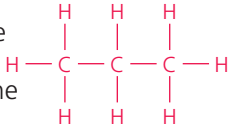
21 A teacher explained the different types of formula used in organic chemistry, using ethene as an example.

Description	Formula
General	C_nH_{2n}
Empirical	CH_2
Molecular	C_2H_4
Structural	

a) Use this example to help you write the formulae described below.

i) The empirical formula of methane
CH_4

ii) The molecular formula of ethane
C_2H_6

iii) The structural formula of propane


b) All the compounds in part (a) belong to the same homologous series and have the same general formula.

i) State the name of this homologous series.
Alkane

ii) State the general formula of this homologous series.
C_nH_{2n+2}

iii) State TWO characteristics of the compounds in a homologous series, other than having the same general formula.

c) All the compounds in part (a) are hydrocarbons. They can undergo complete combustion when burnt in oxygen to form carbon dioxide and water.

i) Write a chemical equation for the complete combustion of propane (C_3H_8).
$C_3H_8(g) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(l)$

ii) Identify ONE solid product and ONE gaseous product that could form during the incomplete combustion of propane. Solid product: carbon
Gaseous product: carbon monoxide

d) When methane reacts with bromine, one of the products is hydrogen bromide.

i) State a condition needed for this reaction.
Ultraviolet light

ii) Name the organic product of the reaction.
Bromomethane

iii) Write a chemical equation for the reaction.
$CH_4(g) + Br_2(g) \longrightarrow CH_3Br(g) + HBr(g)$

(Edexcel IGCSE, Chemistry, Paper 1C, May 2011, 8(a)–(c), (e))

22 This question is about alkanes.
Refer to page T39.

a) The largest use for alkanes is as fuels. However, there are problems associated with the combustion of alkanes, whether complete or incomplete.

i) An incomplete combustion of methane, CH_4, results in the formation of carbon monoxide and water only.

Write the equation for this reaction.

ii) When does incomplete combustion occur?

iii) State TWO problems that result from the incomplete combustion of alkane fuels.

iv) State and explain the main environmental problem arising from the complete combustion of alkane fuels.

b) The reactions of carbon compounds, including alkanes, may be broken down into a series of steps; this is the mechanism for the reaction. The reaction between methane and chlorine may be represented by a mechanism involving three stages — initiation, propagation and termination.

i) Using an electron diagram to show the initiation step of the reaction between methane and chlorine. (Show electrons in the *outermost shells* only.)

ii) Using electron diagrams, give the TWO propagation steps of the reaction between methane and chlorine. (Show electrons in the *outermost shells* only.)

iii) Suggest why a small amount of UV light can result in the formation of a large amount of product.

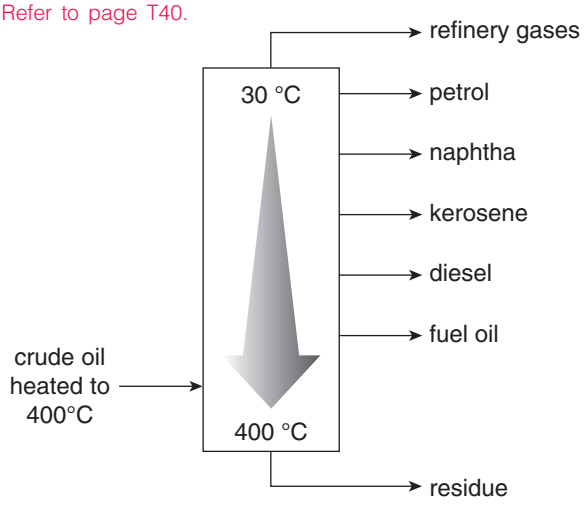
iv) Ethane is a trace product of this reaction. By means of an equation, show how the ethane is formed.

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2011, 7(b)–(c))

21 b) iii) Any two of the following:
- Each member in a homologous series differs from the next by a $-CH_2-$ unit.
- They show a gradual change in physical properties.
- They show similar chemical properties.

23 Crude oil is the source of many useful materials. Crude oil is separated into fractions by fractional distillation.

Refer to page T40.



a) Describe how the naphtha fraction separates from the other fractions.

b) The naphtha fraction is often used to make other useful materials. This involves the cracking of hydrocarbons in the naphtha fraction.

For example:

decane ⟶ hexane + ethene

$C_{10}H_{22} \longrightarrow C_6H_{14} + C_2H_4$

i) Balance the chemical equation given above.

ii) Describe how cracking is carried out.

iii) Why does ethene have different chemical properties from decane and hexane?

(AQA GCSE (Higher Tier), Chemistry, Unit 1, Jan. 2008, 6)

24 Compound W contains carbon, hydrogen and oxygen only. The relative molecular mass of W is 88.0. Complete combustion of 1.32 g of W gives 2.64 g of carbon dioxide and 1.08 g of water.

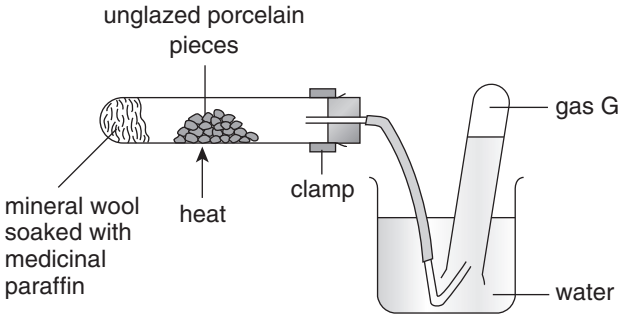
a) Deduce the molecular formula of W.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

b) Given that W has only one functional group, draw TWO possible structures of W.

(HKDSE, Paper 1B, 2013, 3)

25 The following experimental set-up is used to crack medicinal paraffin.



a) What is 'cracking'?

b) What is the purpose of the unglazed porcelain pieces?

c) Why is the mineral wool soaked with medicinal paraffin NOT heated directly?

d) Why should the first tube of gas collected be discarded?

e) Is the gaseous product soluble or insoluble in water? Explain your answer.

f) At the end of the experiment, should the student remove the delivery tube from the water first or should he remove the heating first? Explain your answer.

g) Which one of them, the medicinal paraffin or gas G, has a smaller relative molecular mass? Explain your answer.

h) The cracking reaction in the petroleum industry produces a mixture of products. Suggest a method of separating the mixture.

25 a) Cracking is the breaking down of larger hydrocarbon molecules with heat or catalyst to produce smaller hydrocarbon molecules.
b) As a catalyst
c) The paraffin might vaporize and leave the boiling tube too quickly without undergoing the cracking process.
d) Because it is mainly air.
e) The gaseous product is insoluble in water because the product is collected in the test tube by displacement of water.
f) Remove the delivery tube from the water first.
To prevent water from sucking back.
g) Medicinal paraffin is a liquid while gas G is a gas.
The boiling point of a carbon compound depends on the strength of its intermolecular attractions.
The boiling point of gas G is lower than those of hydrocarbons in medicinal paraffin.
Thus, the van der Waals' forces in gas G is less strong.
Van der Waals' forces increase with relative molecular mass.
It can be deduced that gas G has a lower relative molecular mass.
h) Fractional distillation

26 Alkanes are saturated hydrocarbons which can be obtained from crude oil. Pentane is an example of an alkane. A molecule of pentane contains five carbon atoms.

Refer to page T40.

a) i) State the meaning of the term 'saturated' and of the term 'hydrocarbon' as applied to alkanes.

ii) Give the general formula for the alkanes.

b) Pentane burns completely in oxygen.

i) Write an equation for this reaction.

ii) State how the products of this reaction may affect the environment.

c) Give the name of a solid pollutant which may form when pentane burns incompletely in air.

d) One molecule of C_9H_{20} can be cracked to form one molecule of pentane and one other product.

i) Write an equation for this cracking reaction.

ii) Suggest a type of compound that can be manufactured from the other product of this cracking reaction.

iii) State why a high temperature is needed for cracking reactions to occur.

(AQA GCE Chemistry, Unit 1, Jan. 2010, 4)

27 The hydrocarbons in crude oil can be separated into useful fractions. Refer to page T41.

Fraction	Boiling point range (°C)	Carbon chain length	Relative % in crude oil	Relative % demand
Naphtha	40–170	5–10	10	20
Petrol	40–170	5–10	10	20
Kerosene	170–250	10–14	15	23
Diesel	250–350	14–25	20	25
Fuel oil	370–600	25–70	45	12

a) Why does petrol have a lower boiling point range than fuel oil?

b) Suggest why petrol costs more than fuel oil.

c) Describe how fuel oil can be changed into petrol.

(AQA GCSE (Foundation Tier), Chemistry, Unit 1, Jun. 2008, 8)

28 Crude oil can be separated into different products such as petrol, diesel oil and fuel oil by a process called 'A'. The fuel oil obtained can then be converted into smaller molecules by another process called 'B'.

a) Name process A and process B.

b) i) Explain whether petrol or diesel oil has a higher viscosity.

ii) Explain whether petrol or diesel oil is a cleaner fuel.

c) i) Suggest ONE importance of process B in industry.

ii) One of the compounds in fuel oil is $C_{28}H_{58}$, which can be converted into smaller molecules as shown in the following equation.

$$C_{28}H_{58} \longrightarrow C_{20}H_{42} + 2D$$

(1) Suggest a possible structure of D, and state its systematic name.

(2) Suggest a chemical test to distinguish D from $C_{20}H_{42}$, and state the expected observation.

(HKCEE, Paper 1, 2008, 7)

29 Some simple straight-chain alkenes are shown below:

Refer to page T41.

ethene	$CH_2=CH_2$
propene	$CH_3CH=CH_2$
but-1-ene	$CH_3CH_2CH=CH_2$

a) Give the general formula for alkenes.

b) Write the structural formula of an alkene with the same molecular formula as but-1-ene.

c) Propene was shaken with bromine dissolved in an organic solvent.

i) State the expected observation.

ii) Write a chemical equation for the reaction.

iii) Name the type of reaction that occurred.

d) But-1-ene was shaken with cold acidified dilute potassium permanganate solution.

i) State the expected observation.

ii) Write a chemical equation for the reaction.

iii) Name the type of reaction that occurred.